AKADEMIKA

DCL-CRC

CRC ENCODER / DECODERAND
MSK MODULATION / DEMODULATION
TRAINER KIT

EXPERIMENTAL MANUAL

….A MESSAGE FROM

AKADEMIKA

Today's system designers are faced with tomorrow's problems. ADAVNCED DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.
It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

Company's Motto:

-Light years ahead – refers to leadership.

We strive to be the best in what we do and maintain high standards in the areas of

• Design • Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate, Opp.
Karishma Society, Off Karve Road, Kothrud, Pune,
Maharashtra- 411038
INDIA
Contact No.:+91-7447438443
Email: lab@akademika.in
www.akademika.in

SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.

- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.

- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.

- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.

- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

TECHNICAL SPECIFICATIONS

Clock & Signal Generation
On board Synchronized Carrier.
On board 8 bit variable data pattern.
Manual switch to introduce errors in CRC

Transmitter Section
Data encoding
- Dibit encoder
- CRC

Modulation Techniques
- MSK Modulation.

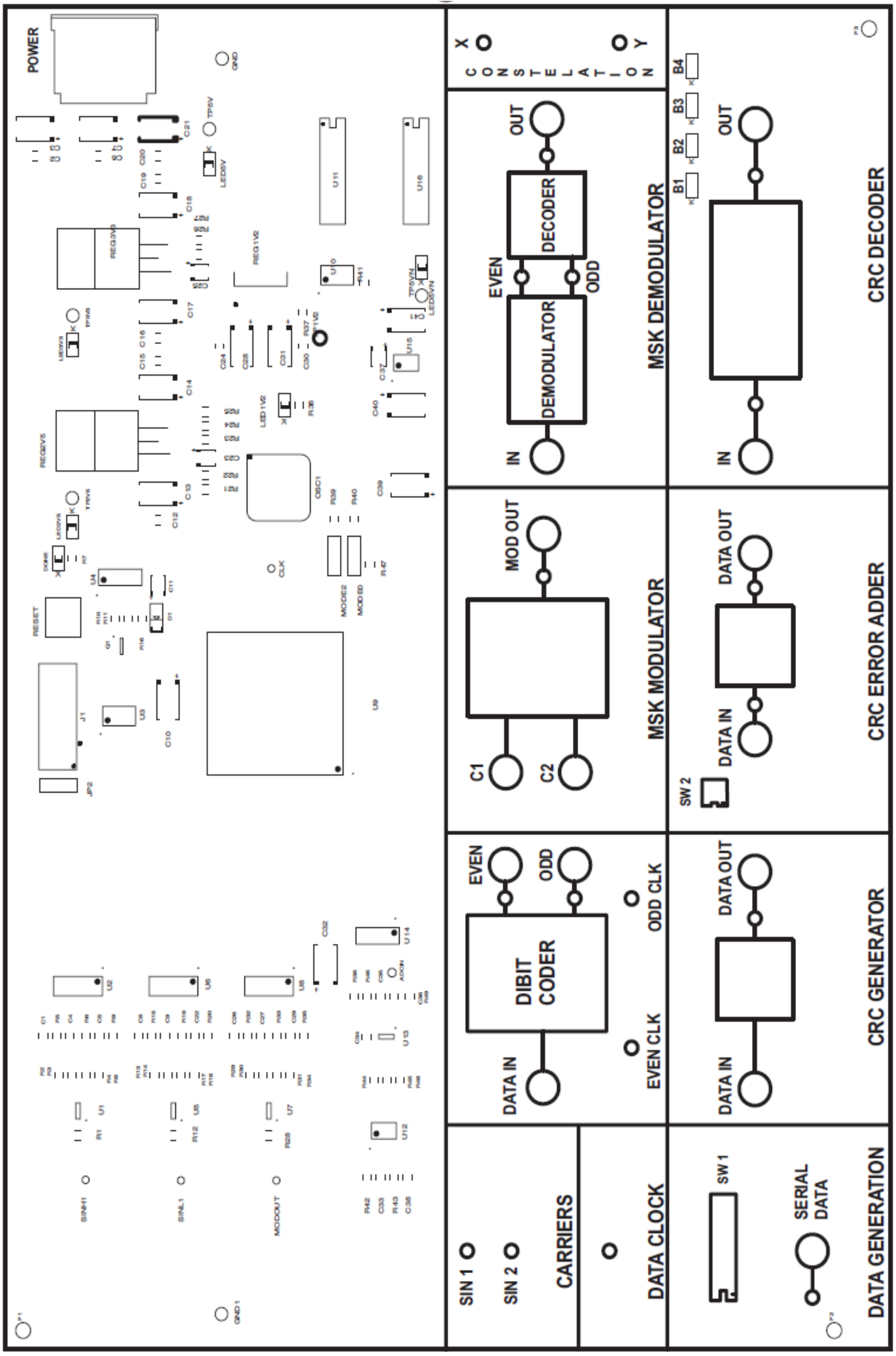
Receiver section
Data Decoding
- Dibit decoder
- CRC decoding
- LED indication for calculated CRC at receiver

Demodulation techniques
- MSK demodulation
- Constellation diagrams for MSK.

Accessories

	DCL-CRC	Quantity
1	Short Links PCS2 Patch cord Male To Male	10
2	DCL-CRC EXPT Manual	1
3	5 PIN DIN CONNECTOR Power Supply	1
4	Power cord	1

FUNCTIONAL BLOCK



INTRODUCTION

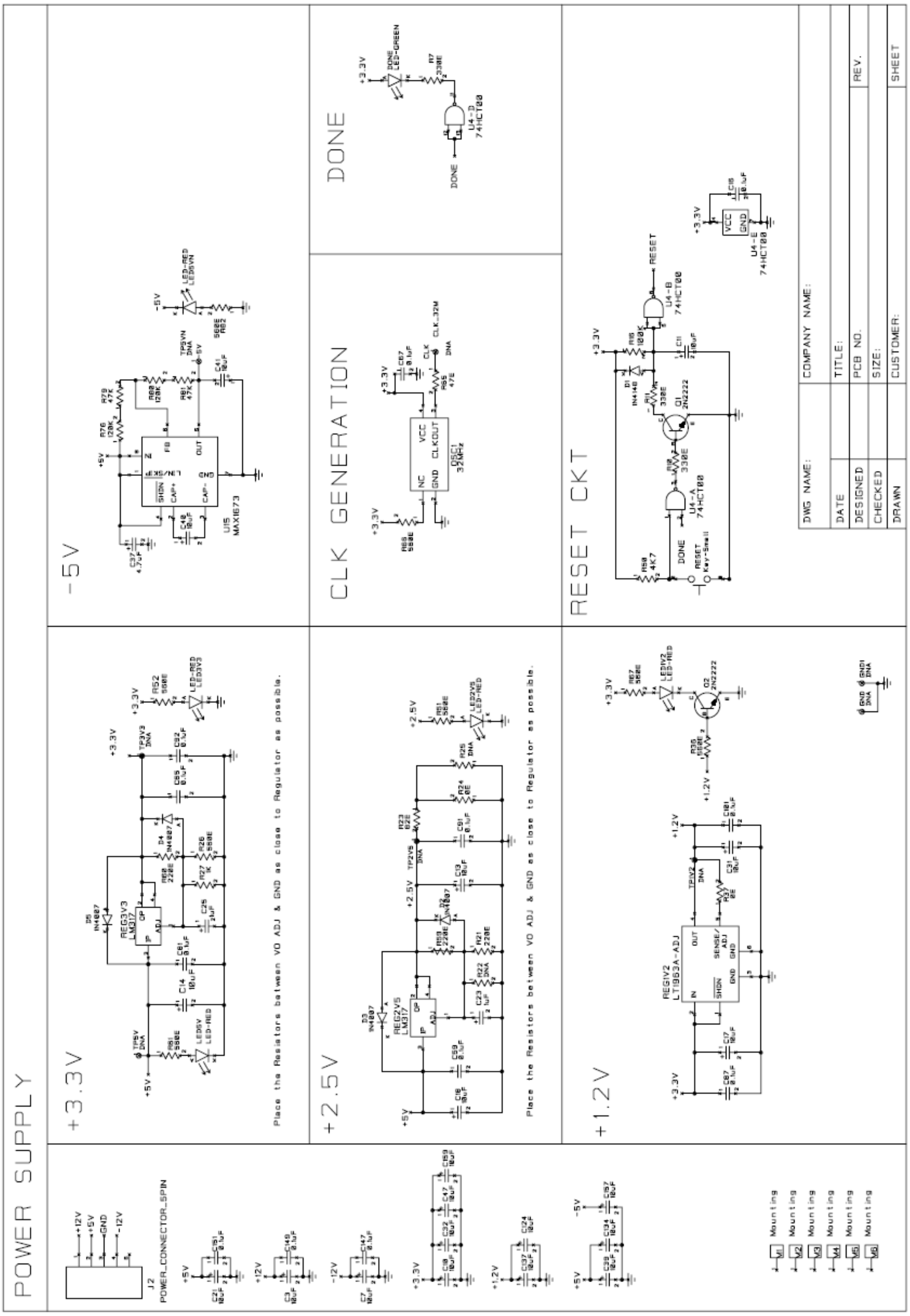
DCL-CRC is a single board digital communication trainer system designed using advanced DDS technology on silicon.

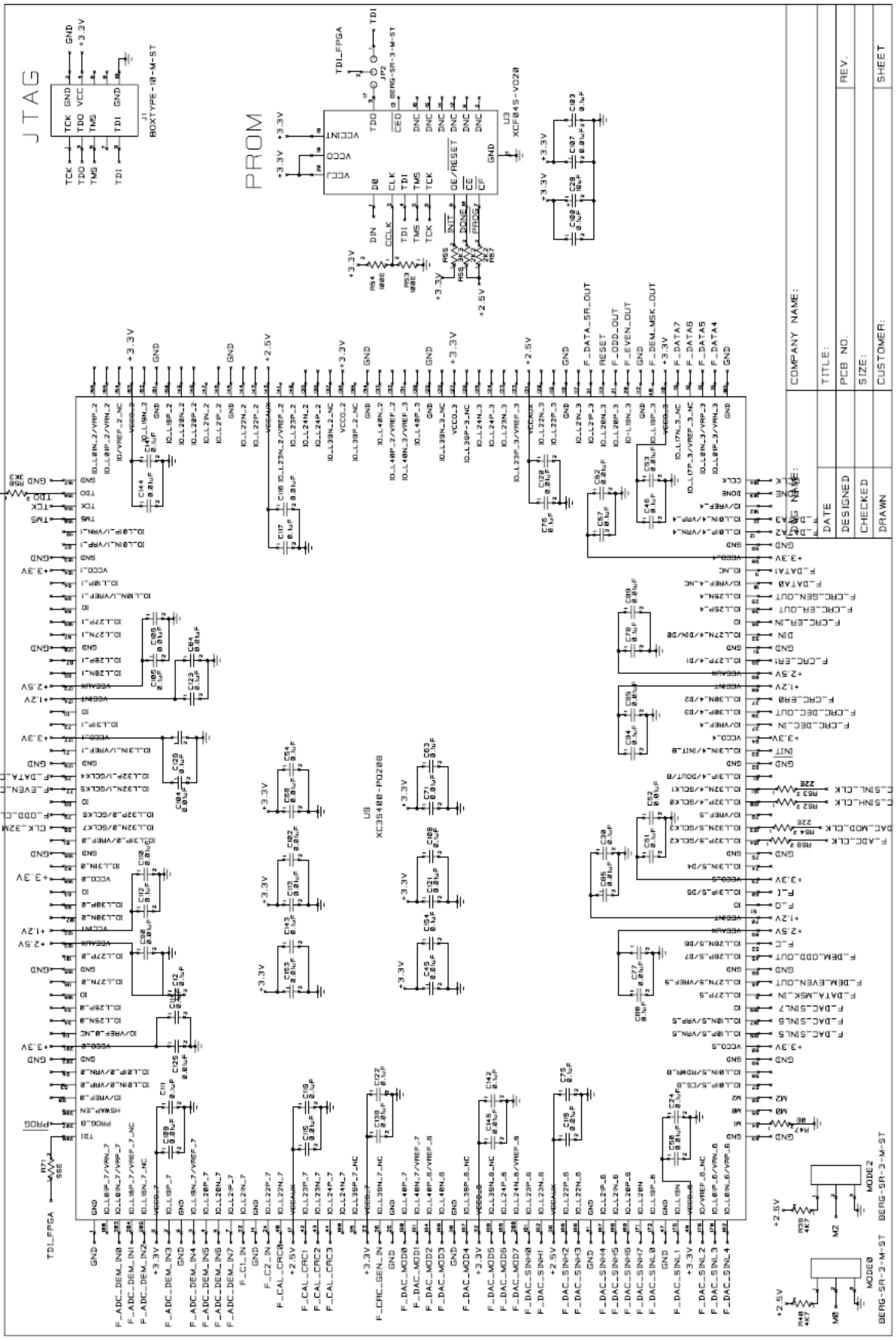
This trainer includes modulation technique like MSK. It also demonstrates the CRC encoding and decoding.

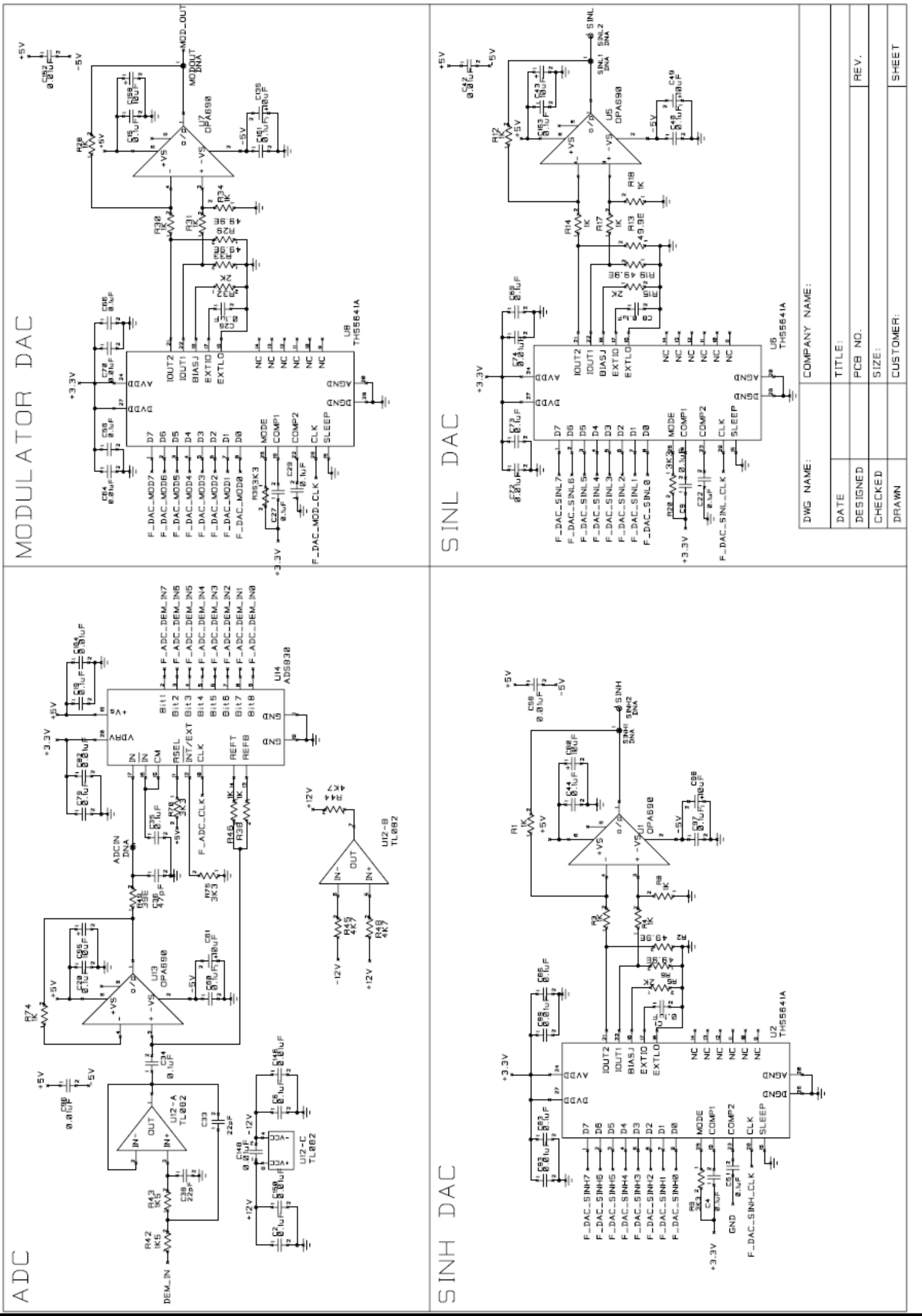
All required input resources are provided on-board. Various test points are provided to observe the intermediate signals.

The LED indications are provided to observe the received CRC. The system comes with descriptive experimental manual.

CIRCUIT DIAGRAM




JTAG
PROM
XCF04S-VO20
U9
XC3S400-PQ208
COMPANY NAME:
TITLE:
PCB NO.
SIZE:
CUSTOMER:
DATE
DESIGNED
CHECKED
DRAWN
REV.
SHEET


MODULATOR DAC
SINL DAC
ADC
SINH DAC

EXPERIMENT NO. 1

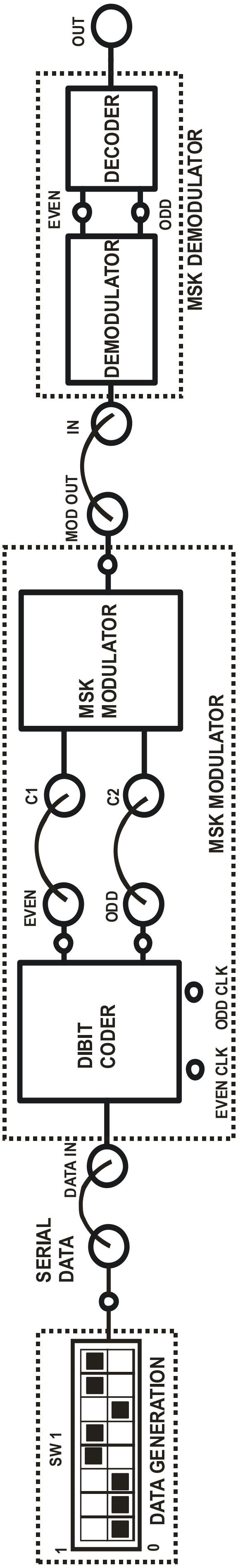


FIG 1.BLOCK DIAGRAM FOR MSK MODULATION AND DEMODULATION

EXPERIMENT NO.1

NAME:

Study of MSK Modulation and Demodulation

OBJECTIVE:

A) To study Minimum Shift Keying modulation and demodulation
B) To study the constellation diagram of MSK.

EQUIPMENTS:

- DCL-CRC Kit
- Power supply
- Connecting Links
- DSO

THEORY:

In digital modulation, **Minimum-Shift Keying (MSK)** is a type of continuous-phase frequency – shift keying that was developed in the late 1960s. Similar to OQPSK, MSK is encoded with bits alternating between quaternary components, with the Q component delayed by half the symbol period. However, instead of square pulses as OQPSK uses, MSK encodes each bit as a half sinusoid. This results in a constant-modulus signal, which reduces problems caused by non-linear distortion. In addition to being viewed as related to OQPSK, MSK can also be viewed as a continuous phase frequency shift keyed (CPFSK) signal with a frequency separation of one-half the bit rate.

Minimum Shift Keying, MSK is a form of FSK (frequency shift keying) or phase shift keying (PSK), MSK uses changes in phase to represent 0's and 1's, with the phase shift used depending on the previous phase value. MSK acts like FSK with minimum difference between the frequencies of the two FSK signals, resulting in a power spectral density that falls off much faster than QPSK (Quadrature phase shift keying) so MSK can operate in a smaller radio bandwidth than QPSK. GSM uses a variant of MSK and QPSK, GMSK (Gaussian MSK)

Minimum frequency-shift keying or minimum-shift keying (MSK) is a particularly spectrally efficient form of coherent FSK. In MSK the difference between the higher and lower frequency is identical to half the bit rate. Consequently, the waveforms used to represent a 0 and 1 bit differ by exactly half a carrier period. This is the smallest FSK modulation index that can be chosen such that the waveforms for 0 and 1 are orthogonal. A variant of MSK called GMSK is used in the GSM mobile phone standard.

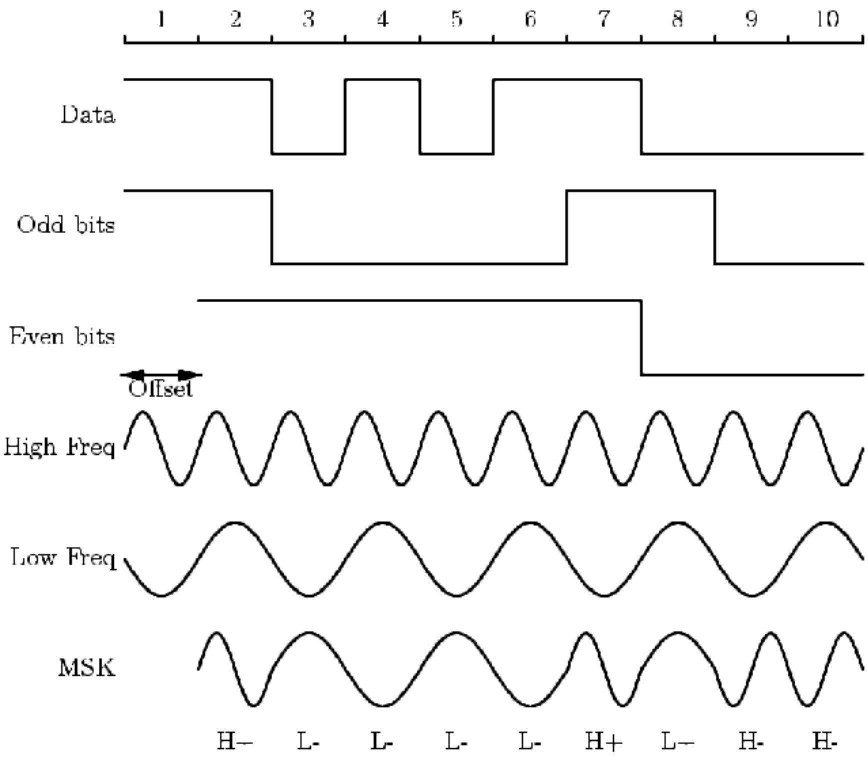


FIGURE 4.3 an example of MSK.

In DCL-CRC the carrier is transmitted according to symbols given in following table.

Sr No	Symbols	Carriers
1	00	HIGH FREQ (180°)
2	01	LOW FREQ (180°)
3	10	LOW FREQ (0°)
4	11	HIGH FREQ (0°)

Frequency of the high frequency carrier = 384.6 KHz.
Frequency of the low frequency carrier = 256.4 KHz.
High frequency (384.6KHz) is 1,5 times the low frequency (256.4KHz).

PROCEDURE:

A) Minimum Shift Keying (MSK) modulation and demodulation

1. Do the Connections as per block diagram shown in fig 1.
2. Connect the power supply to the kit and switch it ON.
3. Set the data pattern as shown in block diagram using **SW1**.Odserve the 8 bit serial data at **SERIAL DATA** post.
4. Observe the carrier used for modulation at **SIN 1** and **SIN2** posts.
5. Connect **SERIAL DATA** to **DATA IN** post of **DIBIT CODER**.
6. Observe **EVEN** and **ODD** data with respect to their clocks.
7. Observe MSK modulated signal at **MOD OUT** post of **CARRIER MODULATOR.**
8. To Demodulate the MSK signal Connect **MOD OUT** to **IN** post of **MSK DEMODULATOR** section.
9. Observe the demodulated **EVEN** and **ODD** signal at **MSK DEMODULATOR** and Compare it with dibit coder.

10. Observe the demodulated signal at **OUT** post of **MSK DEMODULATOR** and Compare it with original signal i.e. with signal at **SERIAL DATA**.

OBSERVATIONS:

- Input Data at **SERIAL DATA**.
- Even and ODD clock signals of Dibit Coder.
- Carrier frequency **SIN 1** to **SIN 2**
- MSK modulated signal at **MOD OUT**.
- Recovered data at **OUT** post of MSK demodulator.

PROCEDURE:

B) Constellation Diagram for MSK

1. Do the Connections as per block diagram shown in fig 1B.
2. Connect the power supply to the kit and switch it ON.
3. Set the data pattern as shown in block diagram using **SW1**.Observe the 8 bit serial data at **SERIAL DATA** post.
4. Observe the carrier used for modulation at **SIN 1** and **SIN2**posts.
5. Connect **SERIAL DATA** to **DATA IN** post of **DIBIT CODER**.
6. Observe **EVEN** and **ODD** data with respect to their clocks.
7. Observe MSK modulated signal at **MOD OUT** post of **CARRIER MODULATOR.**
8. To Demodulate the MSK signal Connect **MOD OUT** to **IN** post of **MSK DEMODULATOR** section.
9. Observe the demodulated **EVEN** and **ODD** signal at **MSK DEMODULATOR** and Compare it with dibit coder.
10. Observe the demodulated signal at **OUT** post of **MSK DEMODULATOR** and Compare it with original signal i.e. with signal at **SERIAL DATA**.
11. To observe the constellation diagram connect **X** to X channel and **Y** to Y channel of CRO. Put oscilloscope display mode on X-Y.

OBSERVATIONS:

- Input Data at **SERIAL DATA**.
- Even and Odd clock signals of Dibit Coder.
- Carrier frequency **SIN 1** to **SIN 2**
- MSK modulated signal at **MOD OUT**.
- Recovered data at **OUT 21** post of MSK demodulator.
- **X** and **Y** post of constellation.

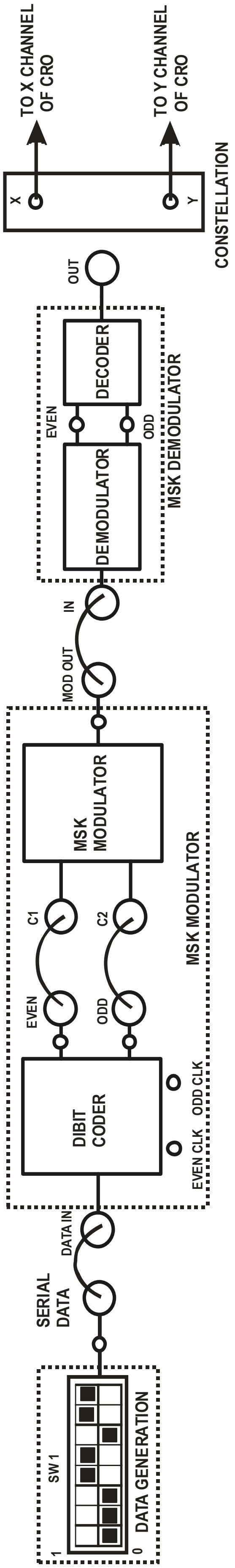
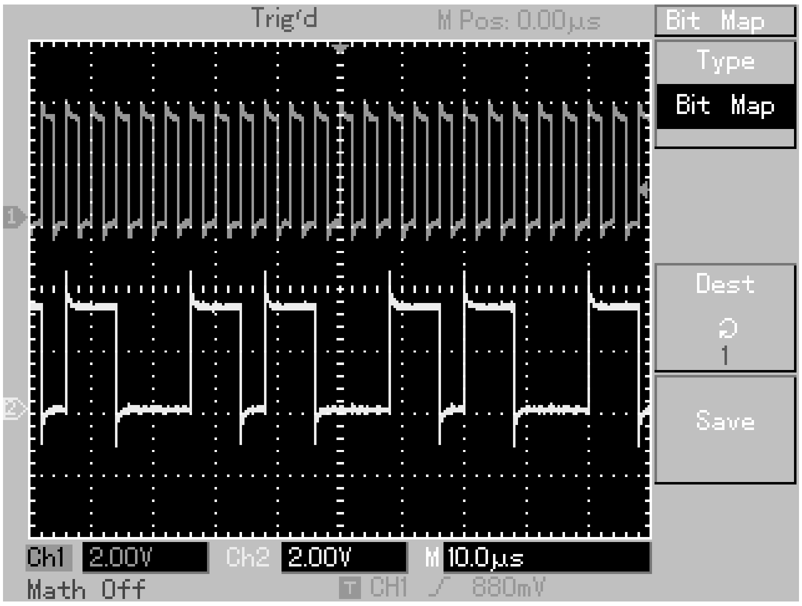


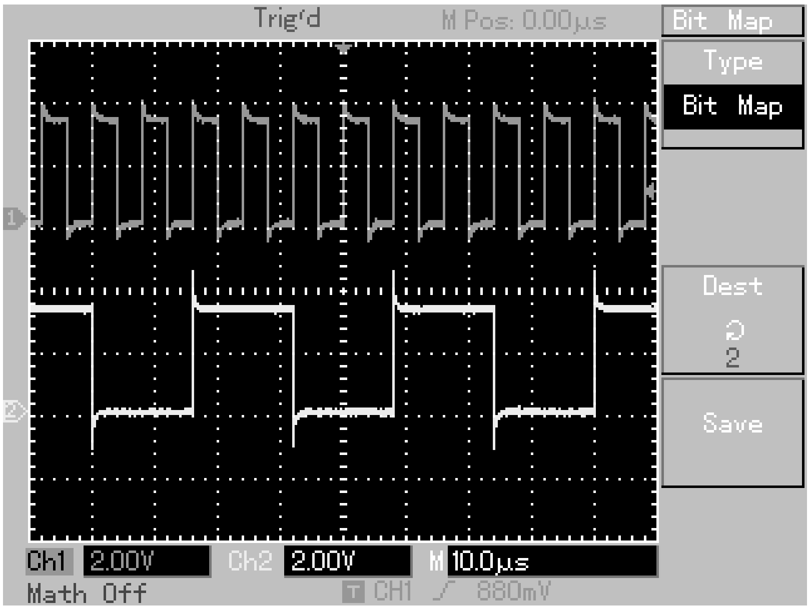
FIG 1B BLOCK DIAGRAM FOR MSK CONSTELLATION

WAVEFORMS:

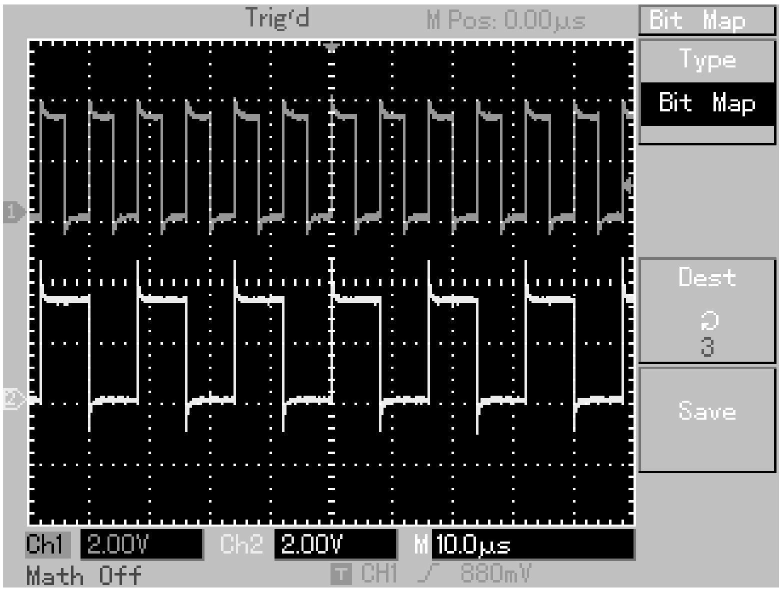
1) CH-1: DATA CLOCK CH-2: SERIAL DATA



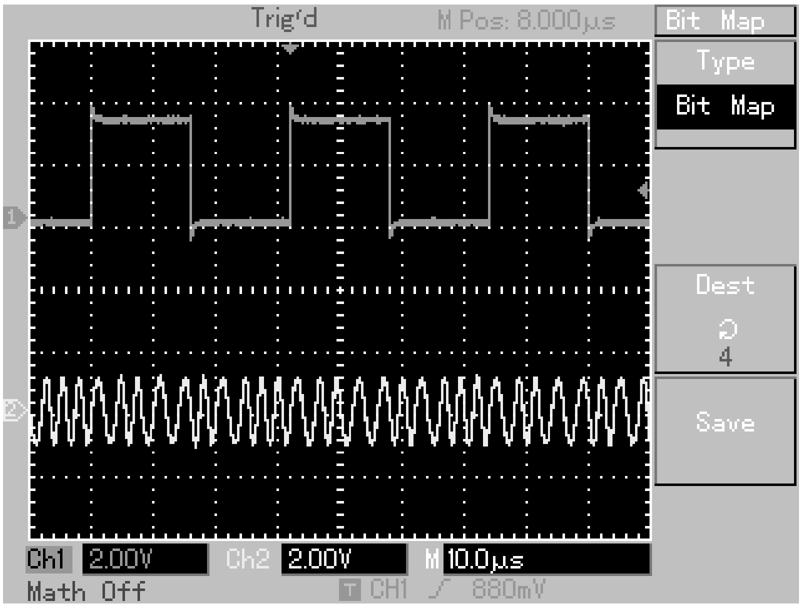
2) CH-1: EVEN CLK CH-2: EVEN DATA



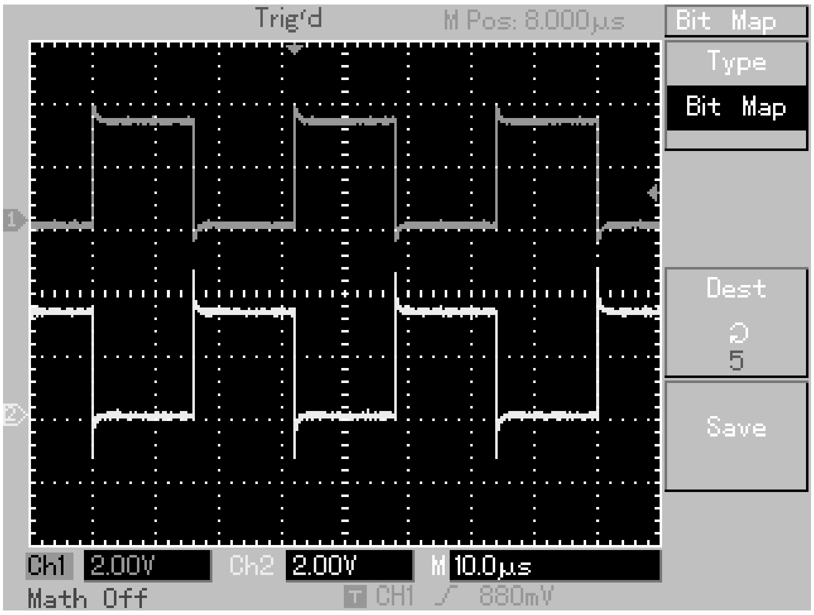
3) CH-1: ODD CLK CH-2: ODD DATA



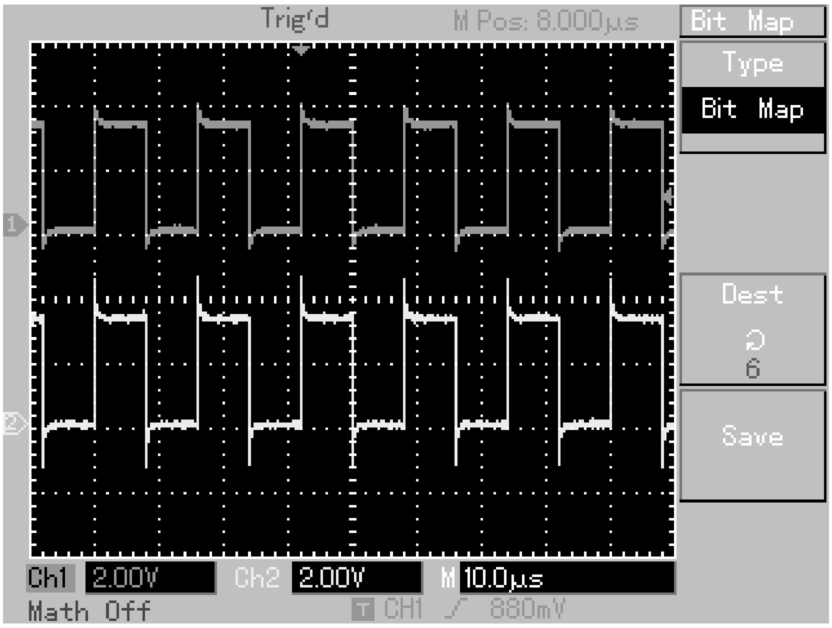
4) CH-1: EVEN DATA CH-2: MOD OUT



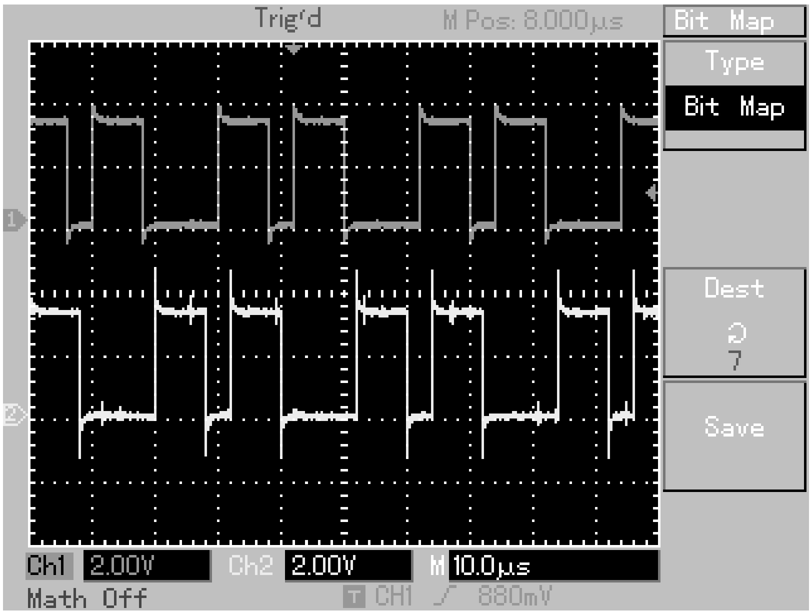
5) CH-1: TX EVEN DATA CH-2: RX EVEN DATA



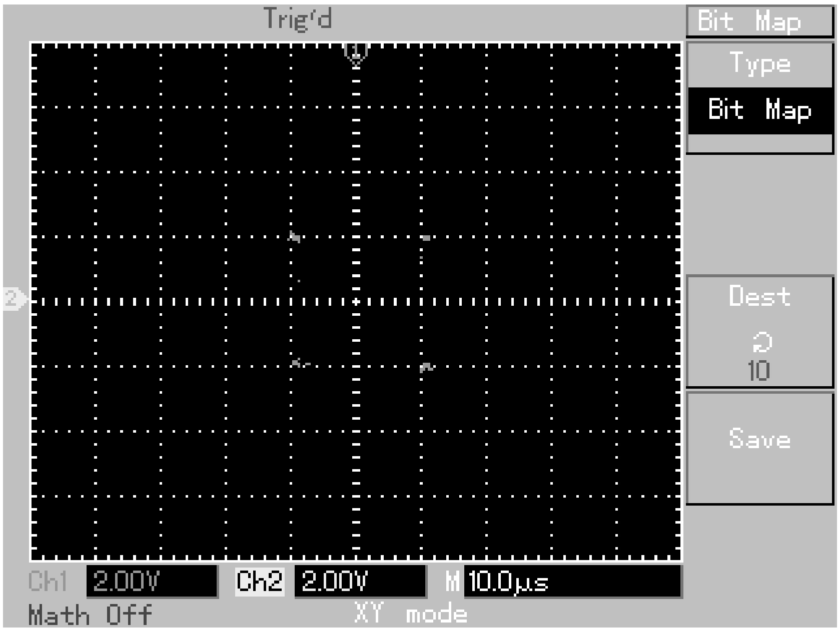
6) CH-1: TX ODD DATA CH-2: RX ODD DATA



7) CH-1: TX SERIAL DATA CH-2: RX MSK DEMODULATOR OUT



8) CONSTELLATION DIAGRAM OF MSK



EXPERIMENT NO. 2

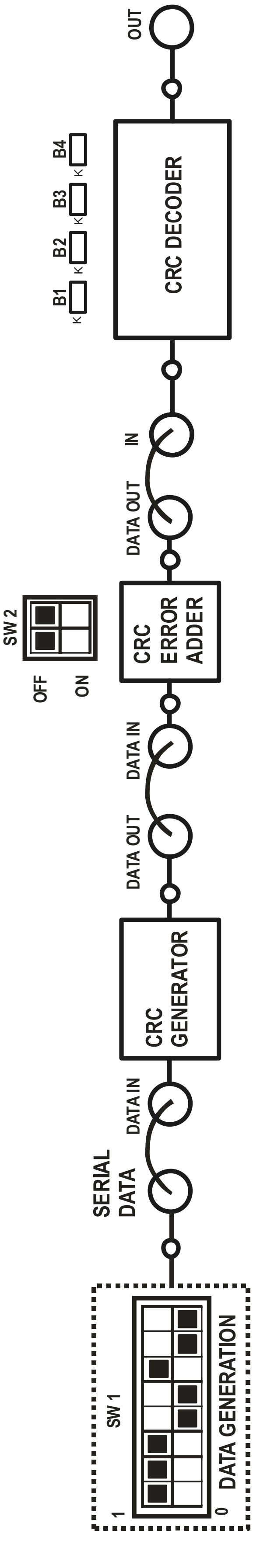


FIG 2.BLOCK DIAGRAM FOR CRC ENCODING AND DECODING

EXPERIMENT NO.2

NAME:

Study of CRC Encoding and Decoding

OBJECTIVE:

To study Cyclic Redundancy code encoding and decoding.

EQUIPMENTS:

- DCL-CRC Kit
- Power supply
- Connecting Links
- DSO

THEORY:

A cyclic redundancy check (CRC) or polynomial code checksum is a non-secure hash function designed to detect accidental changes to raw computer data, and is commonly used in digital networks and storage devices such as hard disk drives. A CRC-enabled device calculates a short, fixed-length binary sequence, known as the CRC code or just CRC, for each block of data and sends or stores them both together. When a block is read or received the device repeats the calculation; if the new CRC does not match the one calculated earlier, then the block contains a data error and the device may take corrective action such as rereading or requesting the block be sent again.
CRCs are so called because the check (data verification) code is a redundancy (it adds zero information) and the algorithm is based on cyclic codes. The term CRC may refer to the check code or to the function that calculates it, which accepts data streams of any length as input but always outputs a fixed-length code. CRCs are popular because they are simple to implement in binary hardware, are easy to analyze mathematically, and are particularly good at detecting common errors caused by noise in transmission channels

Theory of Operation
The theory of a CRC calculation is straight forward. The data is treated by the CRC algorithm as a binary number. This number is divided by another binary number called the polynomial. The rest of the division is the CRC checksum, which is appended to the transmitted message. The receiver divides the message (including the calculated CRC), by the same polynomial the transmitter used. If the result of this division is zero, then the transmission was successful. However, if the result is not equal to zero, an error occurred during the transmission.
The CRC-16 polynomial is shown in Equation 1.below

$$P(x) = x^{16}+x^{15}+x^{2}+1 \qquad \text{-- Equation 1}$$

The polynomial can be translated into a binary value, because the divisor is viewed as a polynomial with binary coefficients. For example, the CRC-16 polynomial translates to 1000000000000101b. All coefficients, like x^2 or x^{15}, are represented by a logical 1 in the binary value.

Example Calculation

In this example calculation, the message is two bytes long. In general, the message can have any length in bytes. Before we can start calculating the CRC value 1,the message has to be augmented by n-bits, where n is the length of the polynomial. The CRC-16 polynomial has a length of 16-bits; therefore, 16-bits have to be augmented to the original message. In this example calculation, the polynomial has a length of 3-bits; therefore, the message has to be extended by three zeros at the end. An example calculation for a CRC is shown below.

Calculation for Generating A CRC

Message = 110101
Polynomial = 101

```
1 1 0 1 0 1 0 0 ÷ 1 0 1 = 1 1 1  0 1
1 0 1
  1 1 1
  1 0 1                 Quotient (has no function in CRC calculation)
    1 0 0
    1 0 1
        1 1 0
        1 0 1
          1 1 0
          1 0 1
            1 1   <-- Remainder = CRC checksum
```

Message with CRC = 1 1 0 1 0 1 1 1

Checking a Message for A CRC Error

Message with CRC = 11010111
Polynomial = 101

```
1 1 0 1 0 1 1 1 ÷ 1 0 1 = 1 1 1  0 1
1 0 1
  1 1 1
  1 0 1              Quotient
    1 0 0
    1 0 1
        1 1 1
        1 0 1
          1 0 1
          1 0 1
            0 0   <-- Checksum is zero, therefore, no transmission error
```

CRC Hardware Implementation

The CRC calculation is realized with a shift register and XOR gates. Figure 10.1 shows a CRC generator for the CRC-16 polynomial. Each bit of the data is shifted into the CRC shift register (Flip-Flops) after being XOR'ed with the CRC's most significant bit.

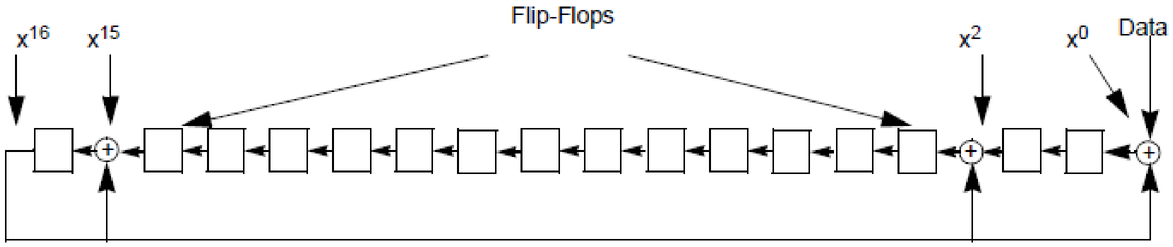


FIGURE 10.1 hardware CRC-16 generator

Example

In DCL-CRC kit the generator polynomial for CRC is X4 + X3 + 1 i. e. (11001).
Select data pattern as 11100100. Internally 4 zeros are appended after actual 8 bit data for transmitting 4 bit CRC. Thus data will be 111001000000.
To Calculate the CRC divide the input data by generator polynomial as shown below.

11100100 0000 ÷ 11001

```
  111001000000 | 11111
  11001
----------------
  0010110
    11001
-----------------
    011110
     11001
--------------------------
     0011100
       11001
----------------------------
        0010100
          11001
------------------------------
          01101 -------- Remainder is nothing but CRC.
```

Thus generated CRC for the data pattern is 1101. And final transmitted data will be (Data + CRC) 11100100**1101.**

In receiver side the same generator polynomial is used to calculate the CRC of the received data. If remainder is zero then received data has no error. CRC can correct single bit error.
In DCL-CRC two errors can be introduced in data at 5th and 8th position using switch SW2. at receiver CRC is being calculated is shown below for data without any errors.

111001001101 ÷ 11001

```
111001001101 | 11111
11001
----------------
0010110
   11001
-----------------
    011110
      11001
-------------------------
      0011111
         11001
---------------------------
         0011001
            11001
-----------------------------
            00000     ------------ Remainder is zero indicated received data has zero errors.
```

If one error is introduced to data using SW2 (left side switch) in 5th position of data then data with error will be received at receiver side. The receiver calculates CRC using same polynomial. If there is some remainder then using look up table the bit with error are found out and simply invert that bit to correct the data.

Data without error at receiver = 111001001101
Data with error in 5th position = 1110**1**1001101

CRC calculation is as shown below;
1110**1**1001101 ÷ 11001

1110**1**1001101 | 11111

```
11001
----------------
0010010
   11001
------------------
   010110
     11001
-----------------
     011111
       11001
-------------------
       0011010
          11001
```

000111

Remainder in this case is 111. The corresponding LED indication is observed on B2, B3, and B4 at CRC DECODER section. The value 111 corresponds to Bit position 5 in look up table and that particular bit is inverted and corrected data at receiver is available which 111001001101 is similarly for another error bit which is in 8th position, the remainder will be 1001 which is indicated on B1 and B4 at CRC DECODER section.

PROCEDURE:

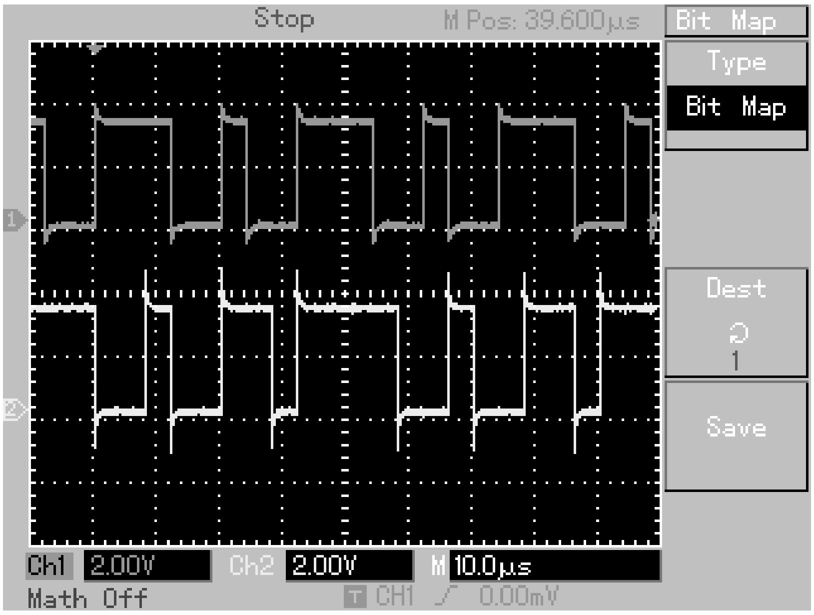
1. Do the Connections as per block diagram shown in fig 2.
2. Connect the power supply to the kit and switch it ON.
3. Set the data pattern as shown in block diagram using **SW1**.Odserve the 8 bit serial data at **SERIAL DATA** post.
4. Connect **SERIAL DATA** to **DATAIN** post of **CRC GENERATOR**.
5. Observe CRC encoded signal at **DATA OUT** post of **CRC GENERATOR**.
6. Connect **DATA OUT** to **DATAIN** post of **CRC ERROR ADDER** block to introduce 2 bit manual error. Introduce error by switch **SW2**.
7. To decode the signal Connect **DATAOUT** to **IN** post of **CRC DECODER** block.
8. Observe CRC decoded and corrected signal at **OUT** post of **CRC DECODER.** Calculated CRC at receiver end is displayed on led **B1** to **B4**.

OBSERVATIONS:

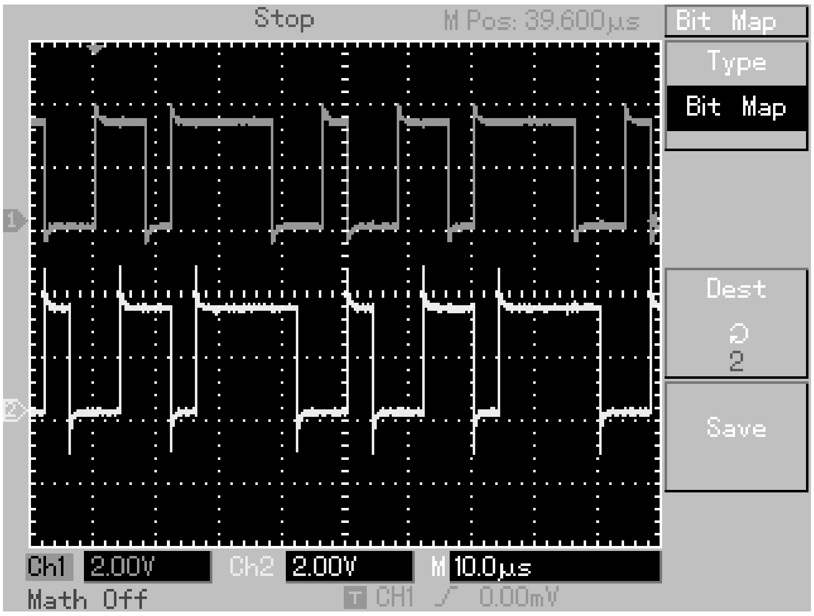
- Input data at **SERIAL DATA** post of data generator.
- CRC encoded data at **DATA OUT** post of CRC generator.
- CRC data with error at **DATA OUT** post of CRC error adder.
- Calculated CRC at receiver on LED **B1 to B4**.

WAVEFORMS:

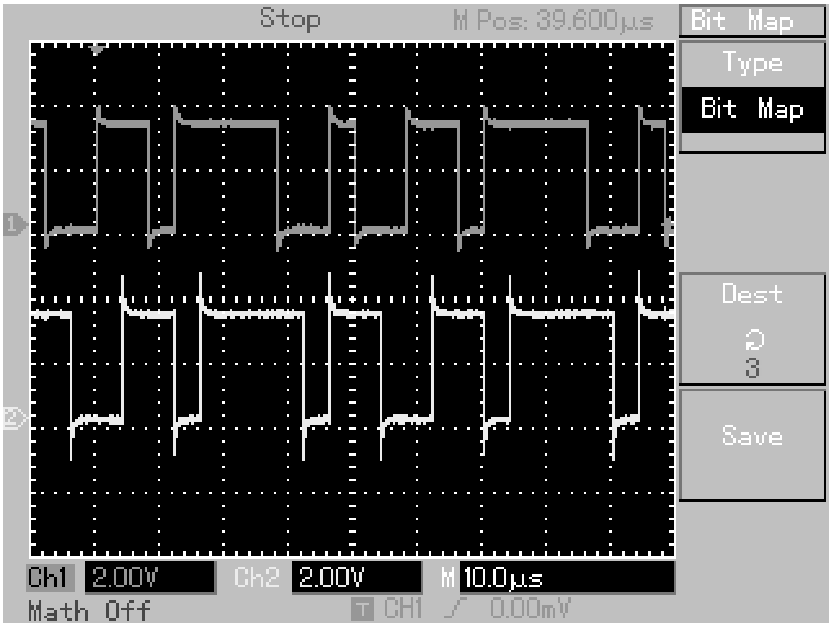
1) CH-1:SERIAL DATA
CH-2: DATA OUT OF CRC GENERATOR



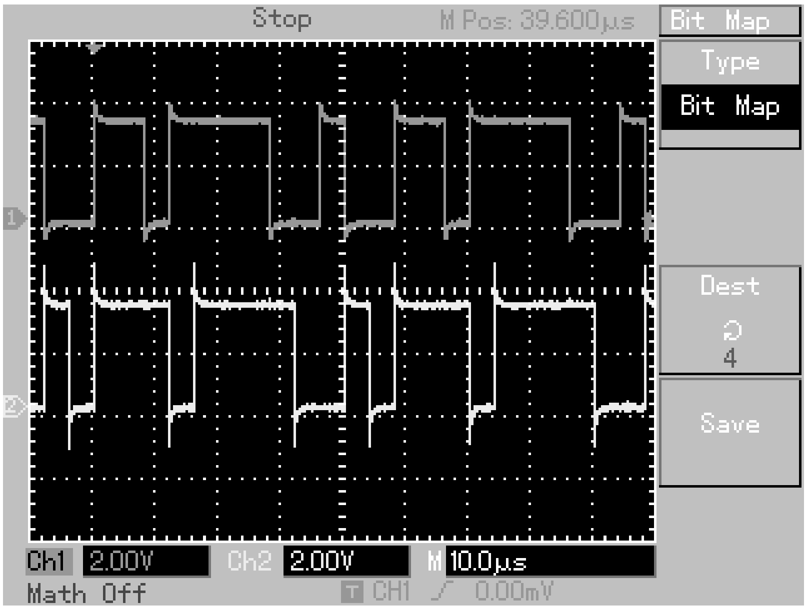
2) CH-1: DATA OUT OF CRC GENERATOR
CH-2: DATA OUT OF CRC ERROR ADDER WITHOUT ANY ERROR.



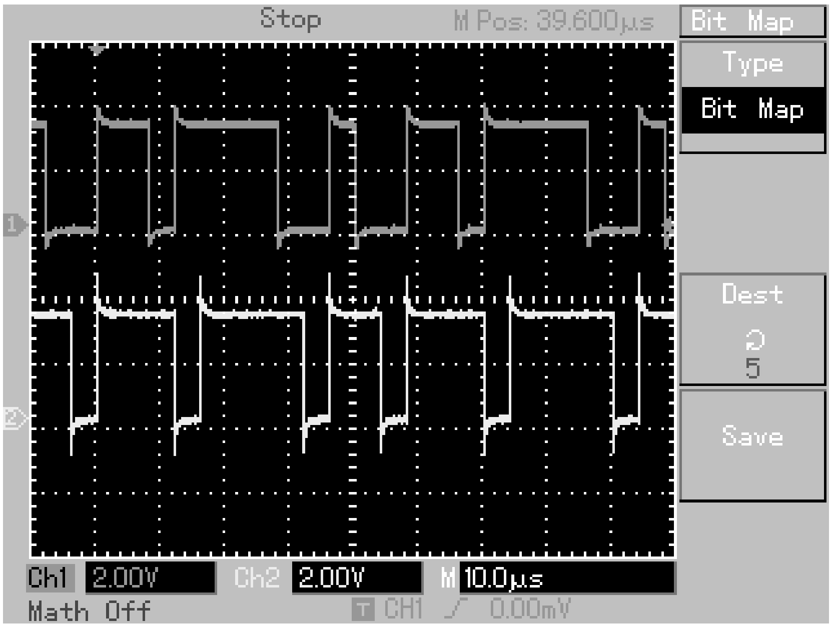
3) CH-1: DATA OUT OF CRC GENERATOR
CH-2: DATA OUT OF CRC ERROR ADDER WITH SW2 FIRST BIT ON



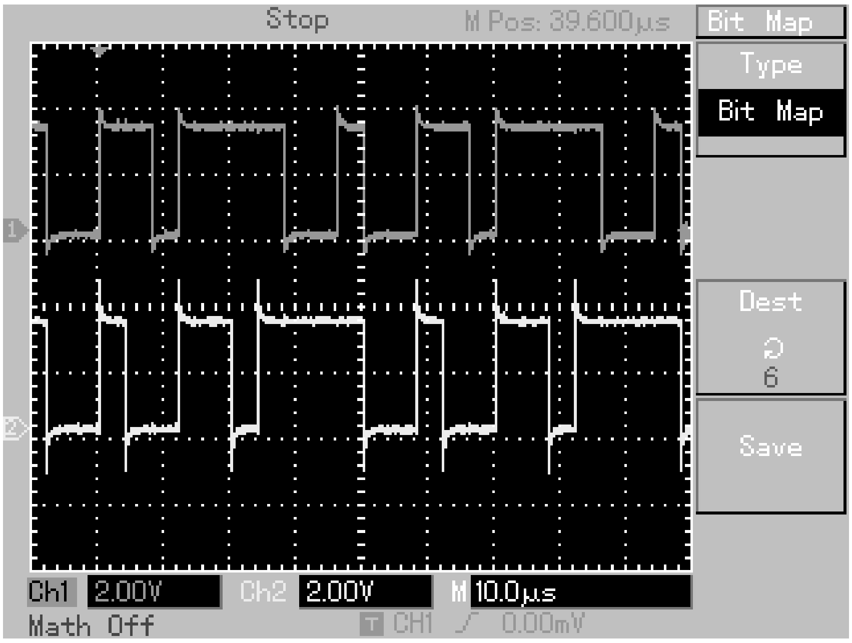
4) CH-1: DATA OUT OF CRC GENERATOR
CH-2: DATA OUT OF CRC ERROR ADDER WITH SW2 SECOND BIT ON.



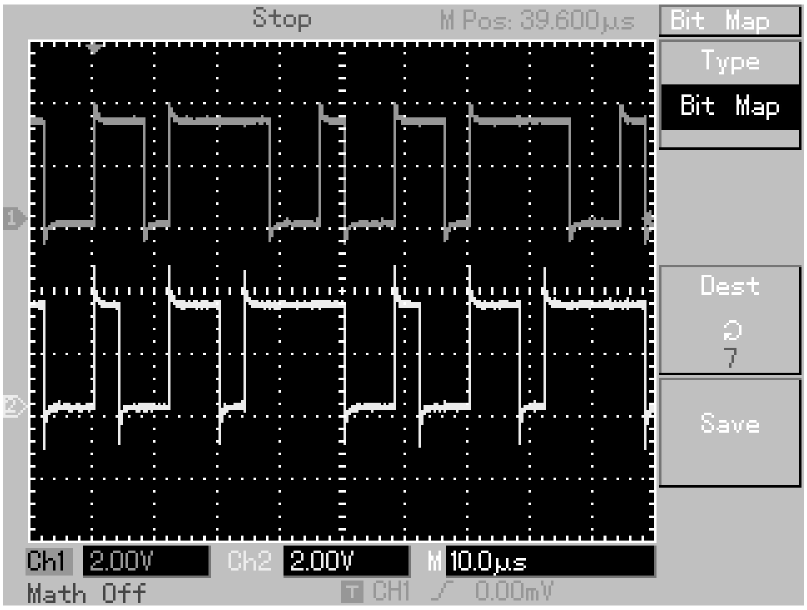
5) CH-1: : DATA OUT OF CRC GENERATOR
CH-2: DATA OUT OF CRC ERROR ADDERWITH SW2 BOTH BIT ON



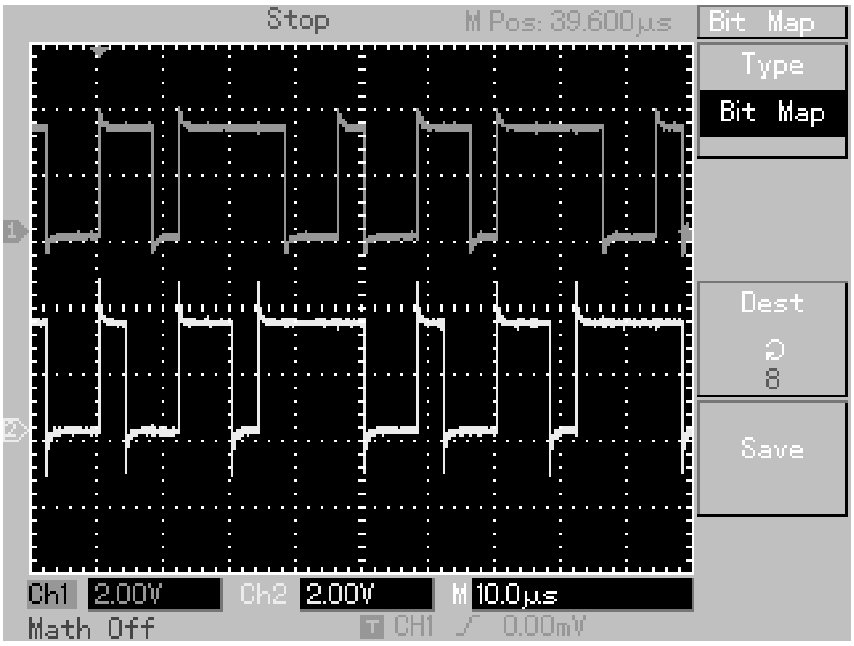
6) CH-1: DATA OUT OF CRC GENERATOR
CH-2: OUT OF CRC DECODER WITHOUT ANY ERROR ADDED.



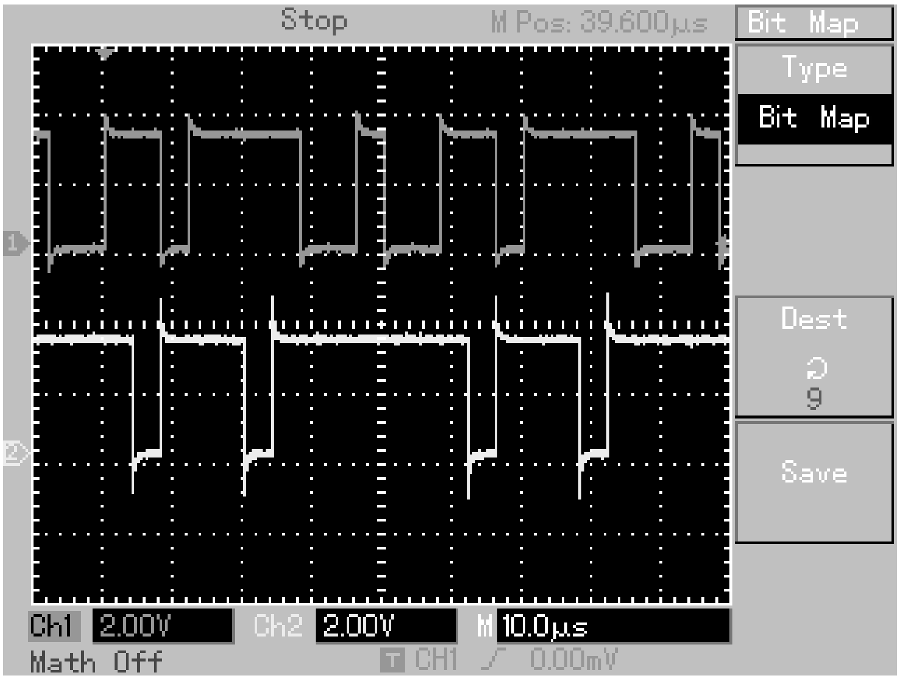
6) CH-1: DATA OUT OF CRC GENERATOR
CH-2: OUT OF CRC DECODER WITH ONE BIT ERROR ADDED.



7) CH-1: DATA OUT OF CRC GENERATOR
CH-2: OUT OF CRC DECODER WITH ANOTHER ONE BIT ERROR ADDED.



8) CH-1: DATA OUT OF CRC GENERATOR
CH-2: OUT OF CRC DECODER WITH TWO BIT ERROR ADDED.


Stop
M Pos: 39.600µs
Bit Map
Type
Bit Map
Dest
9
Save
Ch1 2.00V
Ch2 2.00V
M 10.0µs
Math Off
CH1 0.00mV